

Genetic Parameters and Selection Responses for Alternative Methane Trait Definitions in Pasture-Based Sheep

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Abstract

Background

Enteric methane from ruminants represents a major challenge for environmentally sustainable livestock production. Genetic selection can deliver long-term and cumulative reductions in emissions; however, methane production is positively correlated with key performance traits such as feed intake, body size and growth. Alternative methane definitions, including ratio- and residual-based definitions, have therefore been proposed. The objectives of this study were to estimate genetic parameters for a range of alternative methane trait definitions in sheep, to characterise the genetic and phenotypic relationships between them and with key production traits, and to examine how each definition behaves as the methane trait in a selection index.

Results

A total of 16,535 methane (CH₄) records from 8,354 sheep across 132 Irish flocks were available for analysis. Alternative methane traits were defined as ratios of methane relative to metabolic body weight (MBW), average daily gain (ADG), muscle mass and rumen volume, and as residuals of methane regressed on these traits. Heritability estimates ranged from 0.12 to 0.46. Daily methane was positively genetically correlated with MBW ($r_g = 0.72$), live weight (0.71) and ADG (0.51). Ratio and residual definitions were negatively genetically correlated with their denominator or adjustment traits, while the residual traits remained strongly genetically correlated with absolute methane ($r_g = 0.78$ –1.00). In a selection index of CH₄, MBW and carbon dioxide (CO₂), a favourable response, reducing methane by 0.61 g day⁻¹ with no decline in MBW or CO₂, was reached from every methane trait once appropriate weights on MBW and CO₂ were included in the breeding goal, so the weighting rather than the trait definition determined the outcome.

Conclusions

Methane production in sheep exhibits meaningful additive genetic variation across alternative trait definitions. Ratio and residual methane traits remained strongly genetically correlated with absolute methane but differed in their genetic relationships with body size and growth, and therefore in the correlated responses expected in production traits. Methane can be reduced within a selection index without undesirable responses in other important traits, but the outcome is determined by how methane and these traits are weighted in the breeding goal rather than by the choice of methane trait definition.

Background

Reducing greenhouse gas emissions from ruminant livestock has become a priority for greenhouse gas mitigation strategies worldwide. Enteric fermentation alone contributes approximately 30% of global anthropogenic methane emissions [1], making it one of the largest and most challenging emission sources to address [2]. Throughout this paper, methane

production and methane emissions refer to enteric methane, produced during rumen fermentation and released predominantly by eructation, and exclude methane from manure management. At the same time, rising global demand for animal-sourced protein places increasing pressure on ruminant production systems to deliver sustained reductions in emissions without compromising productivity [3]. The trade-off between greenhouse gas mitigation and food production is particularly evident in grass-based systems, such as Ireland, where agriculture is the largest single sector source of national greenhouse gas emissions, with methane from enteric fermentation in cattle and sheep the dominant contributing gas [4]. Concurrently, despite this high sectoral methane output, Ireland ranks among the most carbon-efficient regions globally for ruminant production when evaluated on a per-kilogram-of-product basis [5, 6]. This places particular emphasis on mitigation strategies capable of delivering sustained reductions in enteric methane emissions without reducing productivity or altering the fundamental structure of pasture-based systems.

Genetic selection for low-emitting animals has emerged as one of the most promising of these methane mitigation strategies, with moderate heritability reported for methane production [7] and the cumulative, long term nature of genetic gain [8]. The recent development of high-throughput phenotyping technologies, most notably portable accumulation chambers (PACs), has enabled routine, large-scale collection of individual-animal methane phenotypes across both research and commercial flocks [9, 10]. Portable accumulation chambers are portable and low-cost, allowing groups of animals to be measured in about an hour under field conditions [45, 46]. Methane measurements from PACs have been shown to be heritable (0.13–0.20 in grazing sheep) [10, 45, 49] and genetically correlated with respiration chamber (RC) methane ($r_g = 0.62$ – 1.00) [10, 48]. PACs also record carbon dioxide (CO_2), which is correlated with feed intake and so offers a proxy for intake in grazing animals, where it cannot be measured directly [47, 48]. Although PACs provide a short spot measurement taken after feed withdrawal, and so

show lower repeatability than RCs [9, 10, 46], they have been consistently validated for ranking animals on methane output [50, 51]. PAC recordings are highly precise (repeatability > 0.99 against known methane releases) [50], and sire rankings are maintained across grazing and controlled environments [49]. As genetic selection depends on ranking animals correctly rather than on absolute emission values, PACs are well suited to large-scale genetic evaluation of methane in sheep [10, 49]. Methane production has also been shown to have strong (positive) genetic and phenotypic correlations with key production traits such as body weight, growth, and feed intake [11, 12]. As a result, direct selection for reduced absolute methane emissions risks inadvertently constraining animal performance.

The development of alternative methane trait definitions, with alternative referring to traits expressed relative to production or as residuals rather than as daily (absolute) methane alone, enables emissions to be quantified in a way that is less directly tied to an animal's body size and growth. Traits such as methane intensity, methane yield, and residual methane express emissions relative to an animal's metabolic demands or level of performance [12, 13]. These traits can identify animals that emit less methane than expected for their level of productivity and have therefore been proposed as a potential means of mitigating unfavourable genetic correlations between methane production and key production traits [14]. Phenotypic analyses have shown that these alternative methane metrics can differ substantially in their relationships with production traits and with each other [15]. Both ratio and residual methane traits present known theoretical limitations, and, as noted by Lassen and Difford [16], the genetic parameters of these alternative definitions remain underexplored. The present study therefore estimates genetic parameters for alternative methane traits in sheep, including their relationships with each other and with key production traits, and explores how they respond to selection.

Methods

99 Gaseous emissions data

100 A total of 16,535 methane measurements were obtained from 8,354 sheep across 132 research
101 and commercial flocks in Ireland between 2019 and 2025. Methane measurements were
102 recorded using PACs following the protocol described by O'Connor et al. [9]. Animals were
103 removed from feed at least one hour prior to measurement. Each animal was then randomly
104 assigned to one of twelve PAC units. Once the chamber was sealed, gas concentrations were
105 measured at entry and again after a 50-minute accumulation period using an RKI Eagle 2 gas
106 monitor (Weatherall Equipment and Instruments Ltd., UK). Concentration changes for
107 methane (CH₄) and carbon dioxide (CO₂) were converted to daily output (g day⁻¹) using
108 chamber-specific calibration equations.

109 Methane (g day⁻¹) and carbon dioxide production (g day⁻¹) measurements were first screened
110 for outliers using a 1.5 × interquartile range criterion applied separately to both gases (n = 511
111 (3.09%) gaseous records removed). All gaseous records were assigned to a contemporary group
112 of date of measurement, flock, and PAC measurement group (i.e., the same 12 animals
113 measured at the same time-point). Only contemporary groups with at least 5 animals were
114 retained, leaving 15,869 records from 8,185 animals.

115 Measurements were collected across multiple physiological stages, including lambs (both
116 sexes, 63 to 365 days of age; 5,564 records from 3,034 animals), hoggets (nulliparous females
117 aged 365 to 600 days; 1,812 records from 1,271 animals), and mature ewes (females with a
118 recorded lambing event aged 431 days to 10.5 years; 7,828 records from 4,327 animals). A total
119 of 25.40% of animals had multiple methane records both within or across physiological stages.

120 Additional phenotypic data

121 Additional phenotypic measurements were available for animals with methane records; these
122 included live weight, average daily gain (ADG), muscle mass, and rumen volume, and were
123 used to derive alternative methane traits.

Live weight

Live weight was recorded immediately prior to PAC measurement using a Prattley electronic weighing scale (O'Donovan Engineering Co. Ltd., Cork, Ireland) and was subjected to outlier removal ($1.5 \times$ interquartile range criterion). Metabolic body weight (live weight^{0.75}) was also derived for each animal. A total of 15,638 records from 8,002 animals were available for analysis.

Average daily gain

Average daily gain was derived for growing animals only (i.e. animals <660 days of age). All individual animal live weight records were obtained from the national sheep database operated by Sheep Ireland. For each methane measurement, live weights recorded within ± 120 days of the PAC measurement date were considered. Average daily gain surrounding each methane measurement was estimated by fitting a linear regression of live weight on age in days, with the regression slope retained as the estimate of ADG (kg day^{-1}). A total of 4,316 ADG estimates were available from 2,658 growing animals after outlier removal ($1.5 \times$ interquartile range criterion).

Rumen volume and muscle mass

Individual animal computed tomography (CT) scan measurements taken within ± 3 days of methane measurement were also available on a subset of animals. Computed tomography provides in vivo estimates of carcass tissue composition by classifying pixels in cross-sectional X-ray images according to their density, allowing fat, muscle and bone tissues to be quantified non-invasively as described by Clelland et al. [17]. From these scans, estimates of rumen volume (litres) and muscle mass (kg) were available. Following outlier screening ($1.5 \times$ interquartile range criterion), 766 CT measurements were available for both rumen volume (746 animals) and muscle mass (751 animals).

Additional data

Auxiliary data relating to animal age, breed composition, animal heterosis and recombination loss, and birth and rearing type were also available. The breed proportion of each animal for the six breeds with the greatest number of methane phenotypes (i.e., Belclare, Charolais, Cheviot, Suffolk, Lleyn, and Texel) was also calculated, together with the combined proportion of all other or unknown breeds. The general heterosis and recombination loss coefficients for each animal were also included [18, 19]:

$$het = 1 - \sum_{i=1}^n sire_i \times dam_i$$

$$rec = 1 - \sum_{i=1}^n (sire_i^2 + dam_i^2) / 2$$

where $sire_i$ and dam_i are the proportion of breed i in the sire and dam, respectively. For all growing animals the birth litter size (defined as whether the lamb was born as a single, twin, triplet or quadruplet), rearing litter size (defined as whether the lamb was reared as a single, twin or triplet), and dam parity (defined as 1, 2, 3, 4, 5, 6 or ≥ 7) were available. For ewes the birth and rearing type corresponding to the litter born to the ewe for the year of measurement were also available.

Methane trait definitions

A suite of methane and CO₂ traits was constructed (Table 1), comprising absolute gas traits (daily CH₄ and CO₂ production) and two groups of alternative methane traits: ratio-based traits and residual traits. Here, alternative methane traits are any methane trait other than daily (absolute) methane production (CH₄, g day⁻¹): ratio traits divide CH₄ by a production or physiological trait, and residual traits take the residual from regressing CH₄ on one or more of these traits.

Absolute Gas Production

Methane production (CH₄) measured in g day⁻¹ represented the total mass of enteric methane, released via eructation during rumen fermentation, emitted per animal per day as captured by

173 PAC measurement. Carbon dioxide production (CO_2), measured concurrently on the same
174 PAC record and also expressed in g day^{-1} , was also analysed.

175 *Ratio-Based Methane Traits*

176 For all animals, two ratio traits were calculated methane per kg of metabolic body weight,
177 representing methane intensity (CH_4/MBW ; $\text{g day}^{-1} \text{ kg}^{-0.75}$), and the other representing
178 methane as a proportion of total methane and carbon dioxide production (CH_4 ratio). CH_4 ratio
179 was calculated on a mass basis as $\text{CH}_4/(\text{CH}_4+\text{CO}_2)$, with both gases expressed in g day^{-1} . For
180 growing animals only, an additional three ratio traits were also computed: methane per kg of
181 average daily gain (CH_4/ADG ; g kg^{-1}), methane per kg muscle mass (CH_4/MM ; $\text{g day}^{-1} \text{ kg}^{-1}$),
182 and methane per litre rumen volume (CH_4/rumen ; $\text{g day}^{-1} \text{ L}^{-1}$).

183 *Residual Methane Traits*

184 Three residual methane traits (RMT) were derived: methane adjusted for metabolic body
185 weight (RMTMBW), methane adjusted for metabolic body weight and carbon dioxide
186 production (RMTMBW+ CO_2), and methane adjusted for average daily gain (RMTADG;
187 calculated in growing animals only). Residual methane traits were generated by regressing
188 daily methane production (g day^{-1}) on the production trait of interest (metabolic body weight,
189 carbon dioxide production; and where available, average daily gain). The residuals from each
190 regression were extracted and used as the phenotype for the corresponding residual methane
191 trait.

192 *Genetic analysis*

193 Variance components for all alternative methane traits were estimated using pedigree-based
194 animal models fitted in ASReml version 4.2 [20]. Pedigree records were available for 36,449
195 animals (all phenotyped animals plus their recursively traced ancestors) and were used to define
196 the additive genetic relationships among animals; 100% of phenotyped animals were
197 represented in the pedigree, 88.5% had both parents known, the pedigree comprised 1,025

unique sires directly above phenotyped animals with a mean pedigree depth of 17.6 generations, and 73.1% of sires linked two or more contemporary groups.

All methane traits were analysed using the following linear mixed animal model:

$$y = \mu + sex + BR + CL + CV + LY + SU + TX + UN + het + rec + age + BT_g + RT_g + BT_e + RT_e + DP + CG + a + pe + e$$

where y was the phenotypic observation for the methane trait under analysis; μ was the overall mean; sex was the fixed class effect of sex (male or female); the fixed effects for breed proportion corresponding to Belclare (BR), Charolais (CL), Cheviot (CV), Lleyn (LY), Suffolk (SU), Texel (TX), and the combined proportion of all other or unknown breeds (UN), were included as covariates; het was the heterosis coefficient; rec was the recombination loss coefficient; age was the age at measurement (weeks of age); BT_g was the fixed class effect of birth litter size of the growing animal itself (single, twin, triplet or quadruplet); RT_g was the fixed class effect of rearing litter size of the growing animal itself (single, twin or triplet); BT_e was the fixed class effect of birth litter size recorded for the ewe in the year of measurement (single, twin, triplet or quadruplet); RT_e was the fixed class effect of rearing litter size recorded for the ewe in the year of measurement (single, twin or triplet); DP was the fixed class effect of dam parity for growing animals; CG was the contemporary group fixed class effect of flock-date-lot of measurement; a was the random animal additive genetic effect; pe was the non-additive random animal permanent environmental effect; and e was the residual random error term, where $\mathbf{A}$ was the numerator relationship matrix, $\mathbf{I}$ was the identity matrix, σ_a^2 was the direct additive genetic variance, σ_{pe}^2 was the animal permanent environmental variance, and σ_e^2 was the residual variance.

This model described above was used for both univariate and bivariate analyses. Residual variance heterogeneity was examined by contemporary-group mean. Within each data source (i.e., flock or management group), contemporary groups were ranked by their mean CH_4 and

split into thirds (low, medium and high mean CH₄), giving 41 classes, each with its own residual variance. This model was used for CO₂ and for the five methane traits with sufficient data spanning multiple physiological stages (CH₄, CH₄/MBW, CH₄ ratio, RMTMBW, RMTMBW+CO₂), and its estimates are reported for these traits, with heritability (h²) and repeatability (t) calculated for each class and averaged across classes weighted by the number of records. For the four traits restricted to a single physiological stage or the sparse computed-tomography subset (CH₄/ADG, CH₄/MM, CH₄/rumen, RMTADG), a single homogeneous residual variance was used. When CH₄/MM and CH₄/rumen were the dependent variables, the repeated animal effect was omitted from the model due to small numbers of repeated records. The genetic correlations reported below (Tables 3–5) are from a single homogeneous residual variance for every trait. As a genetic correlation is the ratio of the genetic covariance to the product of the genetic standard deviations, differences in scale between groups of animals change the covariance and the standard deviations in the same proportion, so the genetic correlation itself is expected to be largely unaffected by residual heterogeneity [52]. This was tested empirically by refitting the CH₄–CH₄/MBW bivariate model separately in young and mature animals; the genetic correlation was unchanged across the full data and both age groups (rg = 0.85 in each case). Heritability (h²) and repeatability (t) were calculated as:

$$h^2 = \sigma_a^2 / (\sigma_a^2 + \sigma_{pe}^2 + \sigma_e^2)$$

$$t = (\sigma_a^2 + \sigma_{pe}^2) / (\sigma_a^2 + \sigma_{pe}^2 + \sigma_e^2)$$

where σ_a^2 , σ_{pe}^2 and σ_e^2 are as defined above; where the permanent environmental effect was omitted, t is not reported. Genetic and phenotypic correlations were obtained from the corresponding fitted (co)variance components, with standard errors derived by the delta method; computations were implemented in R version 4.5.3 [24]. Fixed-effect significance (Wald F-tests) and diagnostics of pedigree completeness, connectedness, and contemporary-group confounding are reported in Supplementary Material S1.

Selection index analysis

To investigate how methane traits respond to selection, an index was built around traits that are widely recorded and of central importance: MBW, an important measure of body size that underlies several traits in sheep breeding indexes, and CO₂, which is recorded alongside CH₄ at every PAC measurement and serves as a proxy for feed intake [47, 48]. The selection index analysis examined how absolute, ratio and residual methane traits themselves operate within an index. The aim was to understand the behaviour of these traits, not to estimate the response to selection under a national breeding index.

Genetic parameters were taken from a trivariate animal model of CH₄, MBW and CO₂ fitted in ASReml [20] to the same 15,869 records, with the fixed effects of the univariate model, and with animal permanent environmental effects assumed uncorrelated between traits. Phenotypic variances (σ^2_p) are the sums of the additive, permanent environmental and residual components, and are therefore conditional on the fixed effects.

Index weights were derived using Smith–Hazel selection index theory [44, 21], with CH₄, MBW and CO₂ recorded on each selection candidate. The index weights are $b = P^{-1}Ga$, where P and G are the phenotypic and additive genetic (co)variance matrices of the three traits and a contains the weight on each trait in the breeding goal. These goal weights are relative weights, not economic values: they define each methane trait as a combination of CH₄, MBW and CO₂, or are chosen to hold the MBW and CO₂ responses at zero, and only their ratios affect the index and responses. The predicted response in each trait is $\Delta G = i \cdot Gb / \sqrt{b'Pb}$, where i is the selection intensity. Responses are reported per generation at $i = 1$; values for $i = 1.7$ and per year (generation interval of 2.8 years) are given in Supplementary Table S2.

Each methane trait was expressed as a breeding goal with weights on CH₄, MBW and CO₂. Absolute methane placed a weight on CH₄ only. The residual traits deducted MBW and CO₂ from CH₄ using either the genetic regression coefficients (genetic residual; 0.833 per kg^{0.75} and -6.09×10^{-5} per g day⁻¹) or the phenotypic regression coefficients (phenotypic residual,

corresponding to RMTMBW+CO₂; 0.423 per kg^{0.75} and 0.00371 per g day⁻¹). The ratio traits CH₄/MBW and CH₄ ratio (CH₄/(CH₄+CO₂)) were linearised around the trait means using a first-order Taylor series approximation [22, 53, 54], which is equivalent to deducting 0.810 × MBW and 0.0151 × CO₂ from CH₄, respectively. Linearised at the means, CH₄ ratio and CH₄/CO₂ give the same goal direction (0.0151 = mean CH₄ / mean CO₂), so the results for CH₄ ratio apply equally to CH₄/CO₂.

The responses attainable by any index of CH₄, MBW and CO₂ were first mapped (Figure 1), and a favourable region was defined as the set of responses in which CH₄ declined and neither MBW nor CO₂ declined. For each methane trait, the additional goal weights on MBW and CO₂ needed to reach a reference favourable response, in which MBW and CO₂ were held at zero response (Kempthorne and Nordskog [43]; related to the desired-gains approach [41]), were then derived. For each methane trait, these weights and the resulting responses are given in Table 6.

Uncertainty in the index results was quantified by Monte Carlo simulation. The 15 (co)variance parameters of the trivariate model were resampled jointly from a multivariate normal distribution, using their REML estimates and the sampling (co)variance matrix from ASReml, so that both the standard errors of the estimates and the correlations among them were accounted for. For each of 10,000 samples, G and P were rebuilt and the index results recomputed. Results are reported as point estimates with 95% intervals (2.5th and 97.5th percentiles of the samples).

The analysis assumes that genetic parameters are constant across generations, that selection is on the candidate's own records (no relatives' records or genomic information), and that all three traits are recorded on every candidate. Analyses were performed in R version 4.5.3 [24].

Results

298 The mean MBW was 22.10 (SD = 4.92) kg, mean live weight was 62.72 (SD = 18.18) kg, while
 299 mean ADG, muscle mass, and rumen volume were 0.18 (SD = 0.10) kg day⁻¹, 11.7 (SD = 3.24)
 300 kg, and 6.16 (SD = 1.49) L, respectively. Descriptive statistics for methane traits are presented
 301 in Table 1. Mean methane production was 17.9 (SD = 7.56) g day⁻¹, with methane intensity
 302 (CH₄/MBW) averaging 0.81 (SD = 0.29) g day⁻¹ kg^{-0.75}. Mean carbon dioxide production was
 303 1,189.6 (SD = 534.2) g day⁻¹ (range 40.9–2,869.1). Methane expressed relative to average
 304 daily gain (CH₄/ADG) had a mean of 137.9 (SD = 189.3) g kg⁻¹, reflecting the small
 305 denominator in this ratio trait. The mean values for methane traits derived from CT
 306 measurements were 1.44 (SD = 0.48) g day⁻¹ kg⁻¹ for CH₄/MM and 2.69 (SD = 0.88) g day⁻¹
 307 L⁻¹ for CH₄/rumen. Residual methane traits were centred around zero by construction, with
 308 phenotypic standard deviations of 6.53 g day⁻¹ for RMTMBW, 5.32 g day⁻¹ for
 309 RMTMBW+CO₂, and 4.75 g day⁻¹ for RMTADG.

310 **Table 1.** Descriptive statistics for alternative methane trait definitions.

Trait group	Trait	No. records	No. animals	μ (SD)	Range
Absolute	CH ₄	15,869	8,185	17.90 (7.56)	4.01 – 40.31
Absolute	CO ₂	15,869	8,185	1189.64 (534.24)	40.92 – 2869.08
Ratio	CH ₄ /MBW	15,638	8,002	0.81 (0.29)	0.14 – 2.38
	CH ₄ ratio	15,869	8,185	0.02 (0.01)	0.00 – 0.19
	CH ₄ /ADG	4,316	2,658	137.89 (189.27)	12.03 – 2058.33
	CH ₄ /MM	766	751	1.44 (0.48)	0.33 – 3.08
	CH ₄ /rumen	766	746	2.69 (0.88)	0.65 – 5.73
Residual	RMTMBW	15,638	8,002	0.00 (6.53)	-20.48 – 21.33
	RMTMBW+CO	15,638	8,002	0.00 (5.32)	-23.38 – 24.15
	RMTADG	4,316	2,658	0.00 (4.75)	-10.23 – 25.82

311

312 *CH₄ is daily methane output in grams per day; CO₂ is daily carbon dioxide output in grams per day; CH₄/MBW is*
313 *methane intensity, expressed as CH₄ per kg of metabolic body weight; CH₄ ratio is the ratio of methane to methane*
314 *and carbon dioxide production; CH₄/ADG is methane per kg of average daily gain; CH₄/MM is methane per kg*
315 *muscle mass derived from computed tomography; CH₄/rumen is methane per litre rumen volume derived from*
316 *computed tomography; RMTMBW, RMTMBW+CO₂, RMTADG are residual methane traits derived from*
317 *regression of CH₄ on metabolic body weight, metabolic body weight and carbon dioxide, and average daily gain,*
318 *respectively.*

319 Genetic parameters

320 Genetic parameters for the methane and CO₂ trait definitions are presented in Table 2. Genetic
321 standard deviations ranged from 0.06 g day⁻¹ kg^{-0.75} for methane intensity (CH₄/MBW) to
322 35.92 g kg⁻¹ for methane expressed relative to growth rate (CH₄/ADG), with corresponding
323 genetic coefficient of variation (CVa) values of 7.44% and 26.05%, respectively. Methane
324 production (CH₄) had a genetic standard deviation of 1.47 g day⁻¹ and CVa of 8.20%, and
325 carbon dioxide production (CO₂) had a genetic standard deviation of 117.18 g day⁻¹ and CVa
326 of 9.85%. The genetic standard deviation for methane expressed relative to muscle mass and
327 rumen volume were 0.20 g day⁻¹ kg⁻¹ and 0.37 g day⁻¹ L⁻¹ with CVa values of 13.59% and
328 13.83%, respectively. Residual methane traits (RMTMBW, RMTMBW+CO₂, and RMTADG)
329 had genetic standard deviations of 1.29, 1.41, and 1.55 g day⁻¹.

330 Heritability estimates ranged from 0.12 ± 0.03 for CH₄/ADG to 0.46 ± 0.11 for CH₄/MM (Table
331 2). Heritability estimates for CH₄ (0.19), CH₄/MBW (0.16), CH₄ ratio (0.19), RMTMBW
332 (0.16) and RMTMBW+CO₂ (0.17), from the heterogeneous-residual model, and for CH₄/ADG
333 (0.12 ± 0.03) and RMTADG (0.28 ± 0.03) were broadly similar. Carbon dioxide production
334 (CO₂), also from the heterogeneous-residual model, was somewhat more heritable than most
335 methane traits (0.25; repeatability 0.43). Methane expressed relative to muscle mass and rumen
336 volume produced larger heritability estimates (0.46 ± 0.11 and 0.35 ± 0.11), although both were

associated with larger standard errors, consistent with their much smaller sample sizes (766 CT records). Where the permanent environmental variance could be estimated, repeatability ranged from 0.20 (CH₄/MBW) to 0.43 (CO₂).

Table 2. Genetic parameters for methane and CO₂ trait definitions.

Trait group	Trait	σ_a	σ_{pe}	h^2 (SE)	t (SE)	CV _a
Absolute	CH ₄	1.47	0.61	0.19	0.22	8.20%
Absolute	CO ₂	117.18	97.26	0.25	0.43	9.85%
Ratio	CH ₄ /MBW	0.060	0.034	0.16	0.20	7.44%
	CH ₄ ratio	1.63×10 ⁻³	1.39×10 ⁻³	0.19	0.32	–
	CH ₄ /ADG	35.92 (4.73)	44.35 (3.68)	0.12 (0.03)	0.30 (0.02)	26.05%
	CH ₄ /MM	0.20 (0.03)	–	0.46 (0.11)	–	13.59%
	CH ₄ /rumen	0.37 (0.06)	–	0.35 (0.11)	–	13.83%
Residual	RMTMBW	1.29	0.90	0.16	0.24	–
	RMTMBW+	1.41	1.14	0.17	0.28	–
	CO ₂					
	RMTADG	1.55 (0.08)	–	0.28 (0.03)	–	–

σ_a = direct genetic standard deviation; σ_{pe} = animal permanent environmental standard deviation; h^2 = heritability; t = repeatability; CV_a = coefficient of genetic variation. Standard errors in parentheses. Trait abbreviations as in Table 1. For CO₂, CH₄, CH₄/MBW, CH₄ ratio, RMTMBW and RMTMBW+CO₂, estimates are from the model with a heterogeneous residual variance (41 classes of contemporary groups by mean CH₄ within data source); h^2 and t are means of the class-specific values weighted by number of records, and standard errors are not available for these estimates. The remaining traits use a single homogeneous residual variance, as the heterogeneous model was not estimable for them. CV_a is not reported for CH₄ ratio: as a dimensionless mass proportion with an extremely small genetic standard deviation, its coefficient of genetic variation is not meaningfully interpretable on the same footing as the other traits. A permanent environmental effect could not be estimated for CH₄/MM, CH₄/rumen or RMTADG (permanent environmental variance at the zero boundary; no repeatability reported).

Correlations among methane traits

Genetic and phenotypic correlations between daily methane production (CH_4), CO_2 , and each alternative methane trait definition are presented in Table 3. Phenotypically, daily methane was strongly correlated with the size-adjusted definitions -- 0.87 with both CH_4/MBW and CH_4/MM , 0.90 with RMTMBW , 0.79 with $\text{RMTMBW}+\text{CO}_2$ and 0.71 with CH_4/rumen -- and essentially perfectly correlated with RMTADG (1.00), but more weakly with CH_4/ADG (0.31) and CH_4 ratio (0.46).

Genetically, daily methane was strongly correlated with methane intensity (CH_4/MBW ; 0.85 ± 0.02) and CH_4/MM (0.85 ± 0.04), moderately-to-strongly with CH_4/rumen (0.78 ± 0.10) and CH_4 ratio (0.71 ± 0.03), and only moderately with CH_4/ADG (0.38 ± 0.08). Notably, all three residual methane traits were strongly genetically correlated with absolute methane: RMTMBW (0.86 ± 0.01), $\text{RMTMBW}+\text{CO}_2$ (0.78 ± 0.02) and RMTADG (1.00 ± 0.00), the last being genetically indistinguishable from daily methane. Daily methane was also strongly genetically correlated with CO_2 production (0.57 ± 0.04 ; phenotypic correlation 0.35).

Table 3. Genetic and phenotypic correlations between daily methane production (CH_4), CO_2 , and each alternative methane trait definition.

Trait	Genetic correlation with CH_4 (SE)	Phenotypic correlation with CH_4
CO_2	0.57 (0.04)	0.35
CH_4/MBW	0.85 (0.02)	0.87
CH_4 ratio	0.71 (0.03)	0.46
CH_4/ADG	0.38 (0.08)	0.31
CH_4/MM	0.85 (0.04)	0.87
CH_4/rumen	0.78 (0.10)	0.71
RMTMBW	0.86 (0.01)	0.90
$\text{RMTMBW}+\text{CO}_2$	0.78 (0.02)	0.79
RMTADG	1.00 (0.00)	1.00

370 *Phenotypic correlation standard errors were all ≤ 0.02 . Trait abbreviations as in Table 1.*

371 Correlations between methane and production traits

372 Phenotypic and genetic correlations between methane traits and production traits are presented
 373 in Tables 4 and 5, respectively. Phenotypically, CH₄ showed weak to moderate positive
 374 correlations with the production traits examined, including metabolic body weight (MBW; 0.36
 375 ± 0.01), live weight (LW; 0.35 ± 0.01), ADG (0.17 ± 0.02), rumen volume (0.31 ± 0.04), and
 376 muscle mass (0.26 ± 0.04). As expected, several ratio-based methane traits showed negative
 377 phenotypic correlations with the denominator production traits, including methane intensity
 378 (CH₄/MBW) with MBW (-0.10 ± 0.01), CH₄/MM with muscle mass (-0.24 ± 0.04), CH₄/ADG
 379 with ADG itself (-0.25 ± 0.02), and CH₄/rumen with rumen volume (-0.45 ± 0.03). Residual
 380 methane traits showed contrasting patterns, with RMTMBW negatively phenotypically
 381 correlated with MBW (-0.08 ± 0.01), whereas RMTADG showed a positive phenotypic
 382 correlation with its own adjustment trait, ADG (0.12 ± 0.02).

383 **Table 4.** Phenotypic correlations between methane traits and production traits (standard errors
 384 in parentheses).

Trait group	Trait	Metabolic body weight	Live Weight	Average daily gain	Muscle mass	Rumen volume
Absolute	CH ₄	0.36 (0.01)	0.35 (0.01)	0.17 (0.02)	0.26 (0.04)	0.31 (0.04)
Ratio	CH ₄ /MBW	-0.10 (0.01)	-0.10 (0.01)	-0.11 (0.02)	-0.04 (0.04)	0.13 (0.04)
	CH ₄ ratio	-0.06 (0.01)	–	-0.01 (0.02)	-0.06 (0.04)	0.19 (0.04)
	CH ₄ /ADG	-0.00 (0.02)	-0.02 (0.02)	-0.25 (0.02)	0.26 (0.05)	0.20 (0.05)
	CH ₄ /MM	-0.09 (0.05)	-0.06 (0.05)	0.01 (0.05)	-0.24 (0.04)	0.13 (0.04)
	CH ₄ /rumen	–	0.04 (0.05)	0.00 (0.05)	-0.03 (0.04)	-0.45 (0.03)
Residual	RMTMBW	-0.08 (0.01)	-0.16 (0.01)	0.02 (0.02)	-0.13 (0.05)	0.07 (0.05)

	RMTMBW+	-0.02 (0.01)	-0.12 (0.01)	0.02 (0.02)	–	–
	CO ₂					
	RMTADG	0.17 (0.03)	0.47 (0.02)	0.12 (0.02)	0.21 (0.08)	0.21 (0.08)

385

386 *Trait abbreviations as in Table 1.*

387 Genetically, CH₄ showed a moderate-to-strong positive genetic correlation with metabolic
388 body weight (MBW; 0.72 ± 0.03), live weight (0.71 ± 0.03), ADG (0.51 ± 0.07), muscle mass
389 (0.42 ± 0.12), and rumen volume (0.68 ± 0.15). Ratio-based methane traits were generally
390 negatively genetically correlated with the denominator production traits, including methane
391 intensity (CH₄/MBW) with MBW (-0.27 ± 0.04), CH₄/ADG with ADG itself (-0.81 ± 0.08),
392 and CH₄/rumen with rumen volume (-0.14 ± 0.30 , a correlation with a wide confidence
393 interval). Methane ratio showed weak to moderate negative genetic correlations with MBW
394 (-0.21 ± 0.04), ADG (-0.04 ± 0.06), and muscle mass (-0.23 ± 0.19). Among the residual
395 methane traits, RMTMBW's genetic correlation with MBW was -0.28 ± 0.04 , and
396 RMTMBW+CO₂'s with MBW was -0.11 ± 0.04 , whereas RMTADG showed a moderate
397 positive genetic correlation with its own adjustment trait, ADG (0.42 ± 0.07).

398 **Table 5.** Genetic correlations between methane traits and production traits (standard errors in
399 parentheses).

Trait group	Trait	Metabolic body weight	Live weight	Average daily gain	Muscle mass	Rumen volume
Absolute	CH ₄	0.72 (0.03)	0.71 (0.03)	0.51 (0.07)	0.42 (0.12)	0.68 (0.15)
Ratio	CH ₄ /MBW	-0.27 (0.04)	-0.27 (0.03)	-0.31 (0.08)	-0.10 (0.16)	-0.31 (0.22)
	CH ₄ ratio	-0.21 (0.04)	–	-0.04 (0.06)	-0.23 (0.19)	0.13 (0.30)
	CH ₄ /ADG	-0.12 (0.07)	-0.19 (0.08)	-0.81 (0.08)	-0.02 (0.21)	0.06 (0.22)
	CH ₄ /MM	-0.26 (0.14)	-0.21 (0.12)	-0.04 (0.13)	–	0.22 (0.30)
	CH ₄ /rumen	–	0.12 (0.12)	-0.04 (0.12)	0.16 (0.22)	-0.14 (0.30)

Residual	RMTMBW	-0.28 (0.04)	-0.44 (0.06)	0.01 (0.05)	-0.52 (0.48)	0.39 (0.42)
	RMTMBW+	-0.11 (0.04)	-0.36 (0.05)	0.02 (0.05)	–	–
	CO ₂					
	RMTADG	0.31 (0.11)	0.94 (0.07)	0.42 (0.07)	0.37 (0.28)	0.37 (0.28)

Trait abbreviations as in Table 1. A genetic correlation between CH₄/MM and muscle mass could not be estimated.

Selection index analysis

Genetic parameters of CH₄, MBW and CO₂

Genetic parameters from the trivariate analysis of CH₄, MBW and CO₂, used for all index calculations, are given in Supplementary Table S3.

The favourable region

Figure 1 shows every response attainable by an index of CH₄, MBW and CO₂. The favourable region within it, in which CH₄ is reduced without a decline in MBW or CO₂, exists, and along its boundary the correlated CO₂ response was positive (+10.1 to +64.5 g day⁻¹). The region is small, covering about 6.5% (95% interval 4.9–7.8%) of the area of attainable responses in Figure 1, because CH₄ is strongly genetically correlated with both MBW and CO₂. Holding both MBW and CO₂ at zero response (Kempthorne and Nordskog [43]; related to the desired-gains approach [41]) reduced methane by 0.61 g day⁻¹ (95% interval 0.46–0.75), 46% (35–54%) of the largest attainable methane reduction (–1.34 g day⁻¹; –1.46 to –1.22). Holding MBW alone at zero gave a similar reduction (0.63 g day⁻¹) but increased CO₂ by 10.0 g day⁻¹. Within the region, the weighting sets the balance between the three traits: an index that increased MBW by 0.12 kg^{0.75} and CO₂ by 20.7 g day⁻¹ still reduced CH₄ by 0.52 g day⁻¹, and some methane reduction remained possible up to an MBW increase of about 0.65 kg^{0.75} (0.52–0.77), or a CO₂ increase of about 75 g day⁻¹ (63–86) (Figure 1).

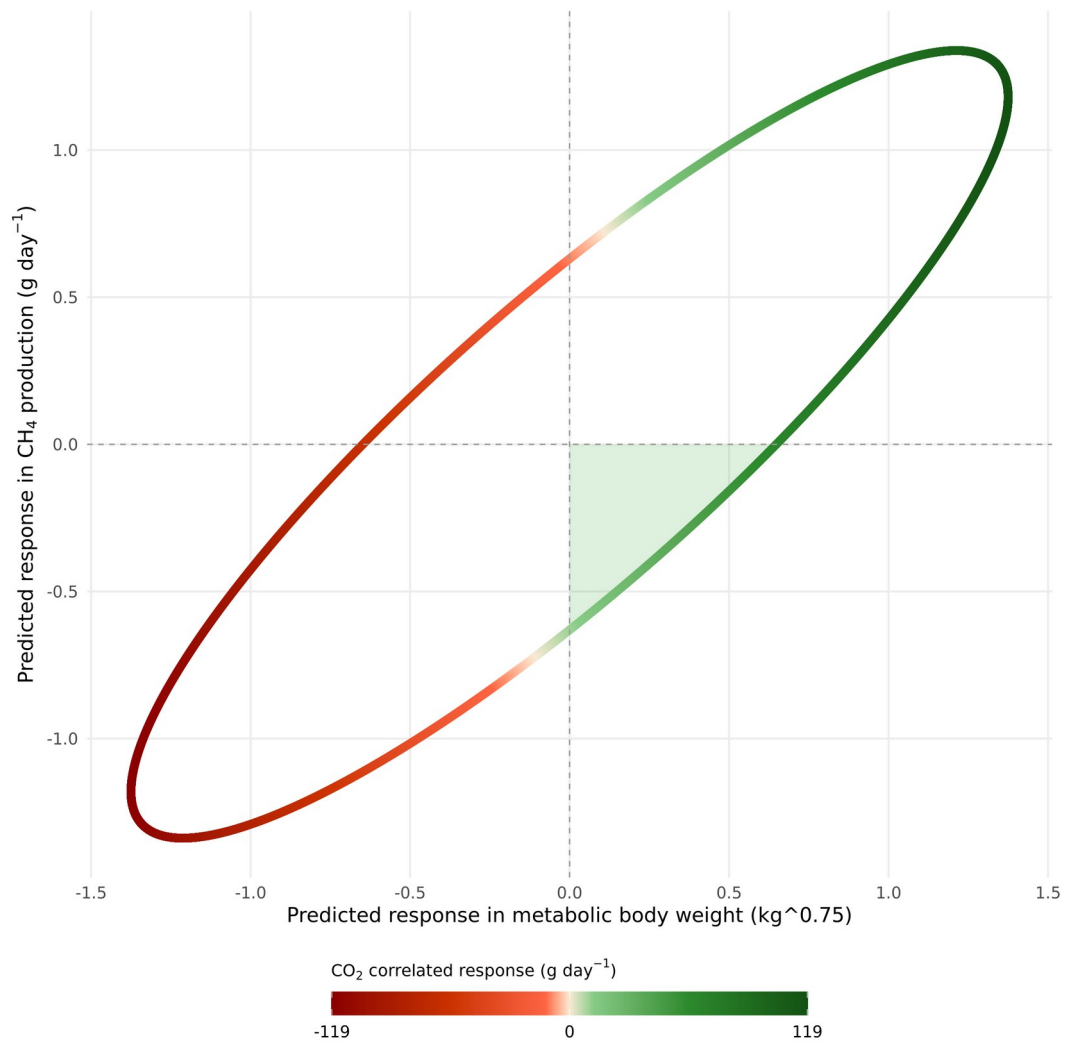


Figure 1. Responses in methane (CH₄) and metabolic body weight (MBW) attainable per generation (selection intensity 1) by any index of CH₄, MBW and CO₂. Colour shows the correlated response in CO₂ (g day⁻¹). The shaded area is the favourable region, in which CH₄ is reduced and neither MBW nor CO₂ is reduced, shown in the CH₄–MBW plane; the CO₂ response along its boundary is positive (+10.1 to +64.5 g day⁻¹). The reference favourable index, with MBW and CO₂ held at zero, lies on the region's MBW = 0 edge at a CH₄ response of -0.61 g day⁻¹.

The reference favourable response (-0.61 g day⁻¹ CH₄, with MBW and CO₂ held at zero) could be reached from a breeding goal built on any of the methane traits, provided appropriate additional weights were placed on MBW and CO₂ (Table 6). All five goals then lead to the

same index and the same responses. The outcome was therefore determined by the weights on MBW and CO₂, not by which methane trait the goal was built on.

Table 6. Additional breeding-goal weights on MBW and CO₂ needed to reach the reference favourable response (CH₄ reduced, MBW and CO₂ held at zero response) when each methane trait forms the breeding goal.

Methane term in the breeding goal	Additional weight on MBW (per kg ^{0.75})	Additional weight on CO ₂ (per g day ⁻¹)	Response: ΔCH ₄ (g day ⁻¹) / ΔMBW (kg ^{0.75}) / ΔCO ₂ (g day ⁻¹)
CH ₄	+1.21	-0.0040	-0.61 / 0 / 0
CH ₄ /MBW (linearised)	+0.40	-0.0040	-0.61 / 0 / 0
CH ₄ ratio (linearised)	+1.21	-0.0191	-0.61 / 0 / 0
Genetic residual	+0.37	-0.0040	-0.61 / 0 / 0
Phenotypic residual (RMTMBW+CO ₂)	+0.78	-0.0078	-0.61 / 0 / 0

Weights are per -1 on the methane term, i.e. the goal is $-(\text{methane term}) + w(\text{MBW}) \cdot \text{MBW} + w(\text{CO}_2) \cdot \text{CO}_2$. They are relative breeding-goal weights (a), not economic values: only their ratios matter, and all five goals give the same index and the same responses. They differ only because each methane term already carries some weight on MBW and CO₂ (ratios linearised around the trait means; residual regression coefficients as in Methods). The corresponding selection index weights (b) are given in Supplementary Table S4.

Discussion

The present study estimated genetic parameters for a comprehensive set of methane trait definitions in pasture-based sheep, comparing absolute, ratio-based, and residual methane traits. Multiple methane traits are used across the literature to quantify emissions, but their statistical properties, and their genetic relationships with production traits, are not always made explicit. Lassen and Difford [16] highlighted the need for further empirical work on the genetic

parameters of ratio- and residual-based methane traits. This study addresses that gap and extends it by examining how these traits respond to selection within a selection index. Heritability estimates across the nine methane trait definitions ranged from 0.12 to 0.46, indicating that methane production carries meaningful additive genetic variation whichever way it is expressed. These estimates lie within the range reported for methane traits in sheep [10, 28], beef [7, 29] and dairy cattle [30, 69, 70], and are comparable in magnitude to traits already used in livestock breeding programmes, such as somatic cell count [31], which has delivered sustained genetic gain in dairy cattle once incorporated into a breeding objective [32].

Feed intake is the main determinant of an animal's daily methane output [55, 56]. Where intake is recorded, methane is therefore commonly expressed per unit of intake (methane yield) or residualised on intake [57, 58]. In grazing sheep, however, feed intake is rarely measured at scale, and scaling or residualising methane against body size, growth or production traits serves as a practical proxy for that missing feed-intake information [12, 13]. The adjustment traits used here are also biologically relevant in their own right. Maintenance energy requirements, and hence much of an animal's feed intake, scale with metabolic body weight, so MBW captures the size-driven component of intake. Average daily gain and muscle mass describe the rate and product of growth, so methane expressed relative to them approximates emissions per unit of output. Rumen volume reflects fermentative capacity and retention time [59]. Carbon dioxide output reflects total metabolic activity and is closely related to feed intake in PAC-measured sheep [47, 48], so CH_4 ratio and $\text{RMTMBW}+\text{CO}_2$ approximate methane relative to intake. The results presented here show both the value and the limits of that approach.

In sheep, the heritability of daily methane matched PAC estimates in New Zealand lambs and ewes (0.19–0.20) [10] and was lower than the respiration-chamber estimate of 0.29 [28]. In cattle, daily methane heritability measures range from 0.15 to 0.42 in beef [7, 29] and from 0.16 to 0.35 in dairy [69, 70], with the highest beef estimates from full-test GreenFeed averages [7,

29], consistent with a single short PAC measurement capturing less of an animal's average emission [9, 46]. The heritability of CH₄ ratio fell within the range of 0.17–0.25 reported by Jonker et al. [10]. Methane relative to product (0.21–0.66) and residual methane (0.38–0.46) were generally more heritable in cattle [29, 69, 70] than the corresponding traits here, which adjust for body size, growth and CO₂ as proxies for intake rather than for measured intake or output.

The present data come from Irish pasture-based flocks of several breeds, recorded at grazing in lambs, hoggets and ewes. Heritabilities for daily methane and CH₄ ratio were similar to those of PAC-measured sheep grazing pasture in New Zealand [10] and Australia [49], and sire rankings for PAC methane have been maintained between grazing and controlled environments [49]. This suggests that the genetic parameters apply to other grazing sheep systems; their transfer to housed or concentrate-fed systems, where diet and intake differ, remains to be tested.

Genetically, the alternative methane definitions were strongly correlated with absolute methane in most cases ($r_g = 0.71$ – 1.00), indicating that they share much of the same underlying additive genetic variation rather than capturing a materially distinct trait. Methane per kg of average daily gain (CH₄/ADG) was the most genetically distinct definition ($r_g = 0.38$ with absolute methane), consistent with the ratio being dominated by its small and highly variable denominator, ADG ($r_g = -0.81$), which is also estimated over a window of up to ± 120 days rather than at the time of the methane measurement. Daily methane was also strongly and positively genetically correlated with body size and growth: metabolic body weight ($r_g = 0.72$), live weight ($r_g = 0.71$), and average daily gain ($r_g = 0.51$). Ratio and residual methane traits did not remove these genetic relationships. Methane intensity and the residual methane traits were negatively genetically correlated with their own denominator or adjustment trait, while remaining strongly genetically correlated with absolute methane itself ($r_g = 0.78$ – 1.00 for the three residual traits). This is consistent with the general statistical behaviour of ratio and

residual variables described by Kennedy et al. [33] and Kronmal [34]. This is expected, as a phenotypic residual is not genetically independent of the traits it is regressed on unless the regression is applied at the genetic level.

Carbon dioxide production was moderately heritable ($h^2 = 0.25$) and strongly genetically correlated with both daily methane ($r_g = 0.57 \pm 0.04$) and metabolic body weight ($r_g = 0.83 \pm 0.02$). CO_2 output reflects total heat production and hence energy intake, and PAC-measured CO_2 is strongly correlated with feed intake in sheep [47, 61] and cattle [60, 62]. This makes it a useful, routinely recordable proxy for intake in grazing animals, which is why it was included in the selection index, although more efficient animals produce less CO_2 per unit of intake [63].

Genetic evaluations can assume a common residual variance across contemporary groups. This assumption is rarely met, and when it is violated, records from more variable groups receive too much weight and variance components and breeding values are biased [64, 65, 66]. It is particularly relevant to PAC methane: between-day and between-group variation in PAC measurements is large [9], emissions differ with site, pasture and protocol [45, 46], and the present data combined animals of very different ages and physiological stages. Classes based on contemporary-group mean methane within flock capture both sources, because mean methane differs strongly with age and stage. Accounting for this heterogeneity lowered the heritability of methane, CH_4/MBW , $RMTMBW$ and $RMTMBW+CO_2$, and of CO_2 , relative to a single pooled residual variance, so the heterogeneous-residual estimates are reported in Table 2 for these traits and for CH_4 ratio. Genetic correlations were little affected: the methane–methane intensity correlation was 0.85 in the full data and in young and mature animals separately, and the young-versus-mature genetic correlation for methane was 0.99. Residual heterogeneity therefore mainly affects heritability estimates rather than the genetic relationships in Tables 3–5, and should be accounted for whenever genetic parameters are estimated from PAC methane recorded across flocks and physiological stages.

Alternative methane traits have been frequently proposed as selection criteria [57, 58, 67, 68], but the central result of the index analysis is that the choice of methane trait is of secondary importance. A favourable region, in which methane is reduced without a decline in body size or CO_2 , does exist, and every methane trait examined, whether absolute, ratio or residual, reached the same response within it once appropriate weights on MBW and CO_2 were included in the breeding goal; all then led to the same index (Table 6). A ratio or residual trait is, in effect, a fixed weighting of methane against its denominator or adjustment trait [33]. It is therefore the weighting of methane, body size and CO_2 in the breeding goal, not the methane trait definition, that determines whether methane can be reduced without reducing size.

This agrees with the observation that direct multi-trait selection can match or outperform selection on a ratio [36, 37, 38], and with the long-standing debate over ratio, residual and index selection for feed efficiency [22, 23]. Within the region, the balance can be shifted towards slightly higher body size or intake, as proxied by CO_2 , at the cost of a smaller methane reduction; the reference index, which holds both at zero response, is the neutral choice. The weights reported here depend on the genetic parameters and are specific to this population.

The weights in Table 6 are breeding-goal weights, not economic values: they are the additional weights on MBW and CO_2 needed to reach a biological target. In practice, the breeding goal would be built from economic values, with methane valued through a carbon price and body size and CO_2 through their effects on output and feed cost [35, 39, 40]. The framework applies unchanged to such a goal: each set of economic values selects one point among the attainable responses in Figure 1.

Several limitations bound these results. Genetic parameters were treated as constant across generations, although they change under selection (Cuyabano et al. [42]). Responses are per generation at $i = 1$; with $i = 1.7$ and a generation interval of 2.8 years they scale by 1.7 and 0.61, respectively (Supplementary Table S2), but realised intensity depends on the breeding

programme, the recording of methane on only part of the population, and the use of genomic information, none of which were modelled. The favourable region is a biological rather than an economic criterion; in practice, the weight given to methane in an index depends on how emissions are priced or rewarded [35, 39, 40]. Finally, MBW is used here as a descriptor of body size, not as a complete representation of productivity, and fecundity, longevity and lamb survival were not included. As most of the feed used by a ewe flock supports ewe maintenance and reproduction rather than lamb growth, these traits are likely to be at least as important as body size and growth when methane is placed in a full bio-economic breeding objective. In addition, only 25.4% of animals had more than one methane record, which limits the precision of the permanent environmental variance and prevented its estimation for the computed-tomography traits and RMTADG.

Conclusions

This study estimated genetic parameters for nine alternative methane trait definitions in a large, pasture-based sheep population, and shows that these definitions differ meaningfully in their genetic parameters and in their genetic relationships with body size and growth. All definitions were heritable, with estimates ranging from low to moderate ($h^2 = 0.12\text{--}0.46$). Ratio and residual methane traits remained strongly genetically correlated with absolute daily methane, showing that they share much of the same underlying genetic variation rather than representing a genetically distinct trait. Daily methane itself was strongly and positively genetically correlated with body size and growth, and scaling or residualising methane redistributed this genetic association rather than removing it. When a methane trait is used on its own, the choice of definition therefore determines the direction and magnitude of the genetic relationship between methane and production.

The selection index analysis shows, however, that this choice matters far less than how methane is weighted against body size and CO_2 . In an index of methane, metabolic body weight and

CO₂, once appropriate weights on body size and CO₂ were included, every trait, whether absolute, ratio or residual, reached the same favourable response and the same index. The weighting of methane, body size and CO₂ in the breeding goal, not the methane trait definition, therefore determines whether methane can be reduced without reducing size. In practice these weights would come from economic values, and the same framework identifies the minimum value that must be placed on methane, relative to body size, for an economic index to achieve this.

Declarations

Ethics approval and consent to participate

Data were generated on growing animals and ewes from 132 Irish sheep flocks. Data collection was approved by the Teagasc Animal Ethics Committee (TAEC2020-252; TAEC2020-258; TAEC0323-374) and the Health Protection Regulatory Authority (AE19132/P112; AE19132/P114; AE19132/P181).

Consent for publication

Not applicable.

Availability of data and materials

The datasets analysed in the current study are not publicly available as they contain commercially sensitive information collected from farmers and private enterprises. Supplementary materials, including full variance components and complete correlation matrices, are available via Zenodo [DOI to be added upon acceptance].

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

[Text — unchanged template placeholder in the submitted docx.]

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Supplementary Material

S1. Fixed-effect significance, pedigree completeness, connectedness and confounding diagnostics.

All 8,185 phenotyped animals were represented in the pedigree; 88.5% had both parents known and 5.3% had neither parent known, and the mean traced pedigree depth was 17.6 generations (median 19, maximum 34). The phenotyped animals had 1,025 unique sires and 5,216 unique dams. Of the 1,025 sires, 73.1% had progeny in two or more contemporary groups and 43.8% in five or more, and the most widely used sire linked 205 contemporary groups.

Confounding between breed proportion and contemporary group was assessed as the within-contemporary-group variance of each breed proportion expressed as a fraction of its total variance, where 1 indicates no confounding and 0 indicates that breed proportion is fully determined by contemporary group. This ratio was 0.64–0.68 for Texel and Suffolk, 0.40 for Belclare, 0.28 for other or unknown breeds, 0.21 for Charolais, 0.15 for Lleyn and 0.04 for Cheviot. The Cheviot and Lleyn proportions were therefore largely nested within contemporary group, and their effects cannot be cleanly separated from those of contemporary group. Both sexes occurred across many contemporary groups, although 82.1% of individual contemporary groups contained a single sex.

Table S1. Significance of fixed effects (Wald F-tests) in the univariate models of the component traits.

Fixed effect	CH ₄	CO ₂	MBW	ADG	Muscle mass	Rumen volume	Live weight
sex	ns	ns	***	***	***	*	***
BR	ns	ns	ns	ns	***	***	ns
CL	***	ns	***	**	***	***	***
CV	ns	ns	**	ns	ns	ns	*
LY	ns	ns	ns	ns	ns	ns	ns
SU	*	ns	***	ns	***	***	***
TX	*	***	***	ns	***	ns	***
Other or unknown breeds	***	***	***	***	***	***	***
het	*	ns	ns	***	***	ns	ns
rec	ns	ns	***	**	***	ns	***
age	*	**	***	***	***	***	***
BTg	***	ns	***	*	ns	ns	***
RTg	***	***	***	ns	***	ns	***
BTe	***	***	***	—	—	—	***
RTe	**	ns	***	—	—	—	***
DP	ns	ns	*	ns	ns	ns	**

* $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; ns, not significant ($P \geq 0.05$); —, term not in the model for that trait. Fixed effects as defined in the model equation (Methods); other or unknown breeds is the proportion of breeds other than the six named, fitted as a covariate. Contemporary group (1,435 levels) was fitted in every model but is not shown, as ASReml omits factors with this many levels from the Wald table by default.

628 **S2.** Predicted responses at selection intensity 1.7, per generation and per year (generation
629 interval 2.8 years). Responses of the reference favourable index (Table 6) multiplied by 1.7 and
630 by $1.7/2.8 = 0.607$, respectively; illustrative values only.

Index	ΔCH_4 (g day ⁻¹) per generation	ΔMBW per generation	ΔCO_2 (g day ⁻¹) per generation	ΔCH_4 per year	ΔMBW per year	ΔCO_2 per year
Reference:	-1.04	0.00	0.0	-0.37	0.00	0.0
CH ₄ reduced, MBW and CO ₂ held at zero						

631

632 **S3.** Genetic parameters from the trivariate analysis of CH₄, MBW and CO₂ used in the selection
633 index analysis.

Trait	Mean	σ_a	σ_p	h^2 (SE)	rg with CH ₄ (SE)	rg with MBW (SE)	rg with CO ₂ (SE)
CH ₄ (g day ⁻¹)	17.90	2.10	4.02	0.273 (0.017)	–	0.69 (0.03)	0.57 (0.03)
MBW (kg ^{0.75})	22.10	1.75	2.30	0.582 (0.019)	0.69 (0.03)	–	0.83 (0.02)
CO ₂ (g day ⁻¹)	1189.64	169.8	265.6	0.409 (0.017)	0.57 (0.03)	0.83 (0.02)	–

634

635 σ_a and σ_p are genetic and phenotypic standard deviations from the fitted variance components; σ_p is conditional
636 on the fixed effects. Phenotypic correlations: CH₄–MBW 0.36, CH₄–CO₂ 0.37, MBW–CO₂ 0.50 ($SE \leq 0.01$). $h^2 =$
637 heritability.

S4. Selection index weights (b, applied to phenotypes; relative weights, not economic values) giving the reference favourable response (CH₄ reduced, MBW and CO₂ held at zero response), written for each methane trait.

Goal coordinates	Weight on methane term (per g day ⁻¹)	Weight on MBW (per kg ^{0.75})	Weight on CO ₂ (per g day ⁻¹)
Absolute CH ₄ goal (CH ₄ , MBW, CO ₂ as themselves)	-0.1618	0.1348	-9.856×10 ⁻⁶
Genetic-residual goal (weights on residual, MBW, CO ₂)	-0.1618	0	0
Phenotypic-residual goal (weights on residual, MBW, CO ₂)	-0.1618	0.0664	-6.095×10 ⁻⁴

The methane term is CH₄, the genetic residual or the phenotypic residual, respectively; all three rows are the same index and give the same responses (Table 6). Weights are selection index weights (b) on the phenotypes, not economic values; multiplying all three by a constant gives the same index and responses. Expressed as relative emphasis (index weight × phenotypic standard deviation from the trivariate model, as a percentage of the summed absolute values), the index places -68% on CH₄, +32% on MBW and -0.3% on CO₂, identically for all three rows. In each row, the MBW and CO₂ weights are those added on top of any already built into the methane term, which is why they are zero for the genetic residual. Weights are on unscaled phenotypes; the CO₂ weight is numerically small because CO₂ is measured in g day⁻¹.

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